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Linux module 2

Assignment 1

1.

Install the OpenSSH server package on your Linux system and verify that the sshd service is running by connecting to localhost using the ssh command.

Ans: **Commands**

sudo apt update

sudo apt install openssh-server -y

Check the SSH service status:

sudo systemctl status ssh

Start and enable the service if it is not running:

sudo systemctl start ssh

sudo systemctl enable ssh

Verify the installation by connecting to localhost:

ssh localhost

When prompted for the first time, type:

yes

Then enter your Linux user password.

Expected Result:

Welcome to Kali GNU/Linux

2.

Generate a new SSH key pair using ssh-keygen, and add your public key to the ~/.ssh/authorized_keys file of your user account to enable key-based authentication.

Ans : **Generate SSH Key Pair**

ssh-keygen

Press **Enter** to accept the default file location and press **Enter** again if you do not want a passphrase.

Add Public Key

cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys

Set proper permissions:

chmod 700 ~/.ssh

chmod 600 ~/.ssh/authorized_keys

Test key-based authentication:

ssh localhost

Expected Result:

SSH logs in using the SSH key without asking for the account password (unless the private key has a passphrase).

3.

Edit the /etc/ssh/sshd_config file to disable password authentication by setting PasswordAuthentication no, then restart the sshd service and test that you can only log in using your SSH key.

Ans : **Edit SSH Configuration**

sudo nano /etc/ssh/sshd_config

Find the line:

#PasswordAuthentication yes

Change it to:

PasswordAuthentication no

Also ensure:

PubkeyAuthentication yes

Save the file (**Ctrl + O**, **Enter**, **Ctrl + X**).

Restart SSH:

sudo systemctl restart ssh

Test SSH login:

ssh localhost

Verify password authentication is disabled:

ssh -o PreferredAuthentications=password localhost

Expected Result:

Permission denied (publickey).

This confirms that only SSH key-based authentication is allowed.

4.

Document each step you took to configure SSH key-based authentication and disable password login, including the exact commands and configuration changes you made.

Hint: Use screenshots or command outputs to support your documentation.

Ans : **Commands Used**

sudo apt update

sudo apt install openssh-server -y

sudo systemctl start ssh

sudo systemctl enable ssh

sudo systemctl status ssh

ssh localhost

ssh-keygen

cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys

chmod 700 ~/.ssh

chmod 600 ~/.ssh/authorized_keys

sudo nano /etc/ssh/sshd_config

sudo systemctl restart ssh

ssh -o PreferredAuthentications=password localhost

Configuration Changes

File Edited:

/etc/ssh/sshd_config

Changed:

PasswordAuthentication yes

to

PasswordAuthentication no

Verified that:

PubkeyAuthentication yes

is enabled.